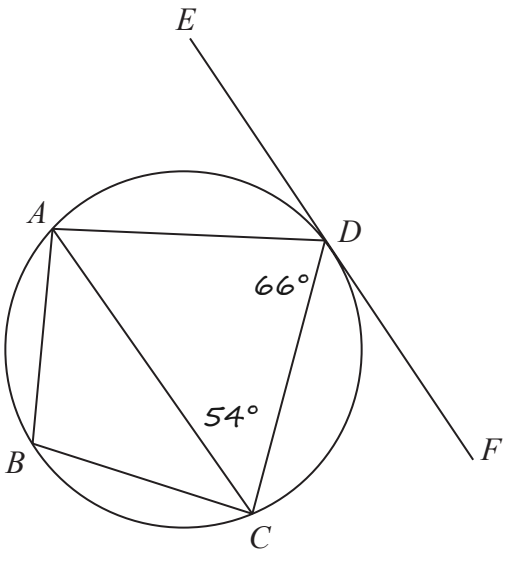
20



A, *B*, *C* and *D* are points on a circle.
EDF is the tangent to the circle at *D*.

Angle *ADE* = 54°
Angle *ABC* = 114°

Work out the size of angle *CAD*.
You must give a reason for each stage of your working.

Angle ACD = 54° (alternate segment theorem).

1 mark
Find54° and66°

Angle ADC = 180° – 114° = 66°

(opposite angles of a cyclic quadrilateral).

2 marks total
Circle reasons linked to their angles

Angle CAD = 180° – 54° – 66° = 60°

(angles in a triangle add to180°).

1 mark
CAD=60°

°

(Total for Question 20 is 4 marks)